A Derek Winchester Joint

Comparison of the BIG 3 Cloud

#init6
518 followers

February 2, 2022

Written by: Derek S. Winchester

Comparison of the Big Three Cloud Platforms for Enterprise

When you need cloud computing technology, you should look at the leaders in cloud infrastructure and solutions. Right now, the three big players in the cloud platform industry are Amazon Web Service (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Each of these platforms has many fans who will argue its merits in tedious detail, explaining why their platform is superior to the others. But no one cloud platform is better in all of its features. (And none is appropriate party talk.) Each of the Big Three has pros and cons. Your enterprise needs will determine which one is the best choice for you. To make the decision easier, we've laid out the significant advantages and disadvantages of each.

AWS

Amazon doesn't run the entire world — yet — but it is influential in many vital areas, including **cloud computing**. AWS is the most widely used globally of the three big players, holding a **34% market share** of cloud hosting providers. However, in the last few quarters, growth has slowed compared to its chief competitors. Still, AWS is an industry leader and offers the following benefits:

- More than 170 featured services.
- A combination of Infrastructure as a Service, Platform as a Service, and Software as a Service.
- More than 55 availability zones in 24 regions.
- Unlimited **storage capacity** – Individual Amazon S3 objects may range from 0 bytes to 5 TB. You may also upload 5 GB at one time.
- Analytics: AWS brings you AWS Elastic MapReduce as its data processing service as well as Data Pipeline, Kinesis Streams, and Kinesis Firehouse: data orchestration, data streaming, and data ingestion tools.

Pricing: AWS has three pricing methods: Pricing for used resources, reserved instance pricing for long-term discounts, and a pay-per-hour model.

AWS claims major clients such as Netflix, BBC, Spotify, Facebook, and LinkedIn.

Disadvantages of AWS

Despite its popularity, AWS has some detractors who point out the **platform's weaknesses**. These include:

- Limits on critical resources, including EC2 distance, EBS volume, and elastic load balancer. However, most companies will never reach these limits. If you need more capability, you can ask for your limits to be expanded.

- Using AWS means investing in education for your staff. The complexity and capabilities of the platform mean that you'll have to spend money on training.

- Additional fees for advanced support. Your monthly fee includes basic support services, but you'll have to cough up more money for help with more complicated issues.

Overall, AWS gets high marks from its clients.

Microsoft Azure

Not surprisingly, **Microsoft Azure** is one of the biggest names in cloud platforms. How big is hard to determine because comparing it directly to CWS is difficult due to corporate structuring and reporting. Microsoft's Intelligent Cloud category includes Azure, but it is not their only holding. Still, Microsoft reports better than predicted growth for 2021, with their Intelligent Cloud division reporting its 2021 first-quarter income as more than $6 billion.

Azure consists of multiple cloud computing services, such as hosted and managed versions of Microsoft products that build and manage cloud applications. Benefits offered by Microsoft Azure include:

- Over 100 services.

- 54 regions in 140 countries – far surpassing that of other cloud providers.

- Storage: Offers Blob Storage, an elastic storage feature that handles unstructured data. Users can upload 5 TB at one time.

- Analytics: Azure includes its Synapse Analytics, which comes with data integration, big data analytics, and enterprise data warehousing. Data tools include HDInsight, Power BI, and Azure Cognitive Services.

- Pricing – Microsoft Azure bills users per minute and rounds up per minute. Some discounts are available for high-volume users.

For companies already using Microsoft products, the transition to Azure is a natural and comfortable move.

Disadvantages of Azure

The **system's complexity** can sometimes lead to employees purchasing additional software that is redundant or simply unnecessary. This platform requires central management of employee spending. Also, some Azure processes require the purchase of stand-alone software that adds significantly to the monthly fees, often through usage charges. The per-minute pricing can give management an unpleasant surprise without careful oversight.

Not everyone finds Microsoft service as helpful as it could be. If you are already dissatisfied with your Microsoft support, Azure may not be for you.

GCP

The last of the big three is Google's offering —GCP. As ubiquitous as the other companies, Google already has its figurative hands in most companies' operations. The Google Cloud Platform unit also includes Google Workspace, so again, direct comparisons to the other platforms are difficult. Still, that unit reported approximately $4 billion in revenues in the first quarter of 2021.

If you can run Google search, Gmail, YouTube, and Google Drive, you should be prepped to manage their cloud platform. Some of the GCP advantages are:

- Over 100 services currently running.
- It offers 25 regions with 76 zones.
- Storage: Includes resumable data uploads and superior interoperability with other cloud storage vehicles. Users can add any object up to 5 TB.
- Analytics: GCP offers BigQuery, its managed enterprise data warehouse for massive datasets, as well as other features such as Cloud Dataflow and Cloud BigTable.

- Pricing: GCP bills per minute but rounds up the final tab per 10 minutes of use.

GCP is not as large as Azure and does not offer as many services as AWS, but many clients are impressed with their unique storage and analytics.

GCP counts PayPal, Toyota, and Twitter as valuable customers of their platform.

Disadvantages of GCP

GCP's reliance on **Google proprietary tech** and its limited choice of programming languages put some users off. Others find that transitioning from GCP to another vendor is unnecessarily complex. Plus, as noted above, it offers fewer services and global data centers than some other platforms.

Final Thoughts

Choosing your cloud platform comes down to your company's needs. Price may be a big consideration for some, and not everyone will like the per-minute billing. Other companies will want the broadest range of services possible and the largest number of regional data centers. You may also be inclined to dance with the companies you already tango with. Microsoft may seem too familiar to pass up.

Container programs are also an important consideration. Deploying Kubernetes is essential to enterprise use of the cloud. Google is the original home of Kubernetes, but all three platforms run the container program, although they may run others as well. Each platform offers excellent features for enterprises, as the companies' lists of big clients would attest to.

Consider your analytics needs and storage capabilities as well. Remember, there is no best platform for everyone. So before choosing your cloud partner, analyze your company's needs and the total cost of signing on with one of the big three, including training. And rest assured that you cannot go too far wrong with any of the triumvirate.